

TRAN VAN NHA

AI Engineer | LLM Agents | Machine Learning

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[linkedin.com/in/nhavan204](https://www.linkedin.com/in/nhavan204) | Hanoi, Vietnam | Open to relocation to Japan

Professional Summary

AI Engineer with **1 year of hands-on experience building agentic workflows** at Viettel Cyber Security, with a focus on production-oriented SOC autonomous. Experienced in multi-step agent workflow design, retrieval-augmented generation (RAG), LLM evaluation/observability, and integration with SIEM, SOAR, EDR, and on-premise systems. Improved automated SOC case handling while maintaining **zero False Negatives** in evaluated workflows. Strong machine learning research background spanning intrusion detection, speaker verification, multimodal learning, and applied AI systems. **JLPT N3**; seeking AI Engineer / LLM Engineer opportunities in Japan.

Professional Experience

AI Engineer – Agentic AI & Workflow Automation 2025 – Present
Viettel Cyber Security *Hanoi, Vietnam* **Project: Agentic AI for SOC Tier 1 Automation**

- Designed and productionized **multi-step LLM-agent workflows** for SOC Tier 1 alert investigation, covering data preprocessing, process-tree analysis, command-line investigation, evidence validation, contextual reasoning, and final security classification.
- Used **QwenPaw** to prototype and evaluate agent execution patterns, tool integration, and local agent workflows, then adapted validated approaches to SOC automation use cases.
- Engineered robust **structured-output pipelines, workflow guardrails, and fallback mechanisms** to handle missing fields, malformed model responses, long-context inputs, and heterogeneous security telemetry.
- Integrated agentic AI workflows with **SIEM, SOAR, EDR, REST APIs, and on-premise infrastructure**, addressing data-schema, API, and deployment compatibility across different customer environments.
- Improved automatic case-closure rates compared to handling execute from approximately **35% to 80%** for Linux alerts and **40% to 93%** for Windows alerts while maintaining **FN = 0** during evaluation.
- Increased successful automated analysis coverage from approximately **85% to 98%** through prompt engineering, input normalization, error handling, and systematic failure-case analysis.
- Used **Langfuse** for LLM tracing, prompt evaluation, observability, failure analysis, and iterative optimization of agent workflows.
- Collaborated with SOC analysts and engineering teams to translate expert investigation procedures and decision logic into **repeatable, production-oriented AI workflows**.

Project: Voice Biometric Authentication & Anti-Spoofing Oct 2025 – May 2026

- Developed a voice biometric authentication pipeline for security-sensitive identity verification using **ECAPA-TDNN** and **ResNet**-based speaker verification models.
- Researched **TimeKAN**-based temporal modeling and integrated speaker verification with anti-spoofing components for replay and synthetic-speech attack robustness.
- Implemented data preprocessing, feature extraction, model training, evaluation, and inference pipelines using **PyTorch** and audio-processing libraries.

Selected AI Projects

AI Assistant for AI Lab – Hierarchical RAG & Database Orchestration Mar 2025 – Jun 2026
AI Lab, Hanoi University of Industry

github.com/vannha2004/AI-Assistant

- Developed an end-to-end **AI Assistant** for the university AI Lab using **RAG** to answer questions from domain-specific knowledge sources.
- Designed a **hierarchical retrieval workflow** that first routes queries to the most relevant knowledge partition, then performs fine-grained document retrieval to reduce search space and improve relevance.
- Implemented document preprocessing, embedding generation, database routing, context retrieval, prompt construction, and LLM response generation as a modular pipeline.
- Added retrieval-result caching to reduce redundant database queries and improve response latency for frequently accessed knowledge.

Safety-First Hierarchical Intrusion Detection System for Edge Environments Jun 2025 – Mar 2026

- Designed a hierarchical intrusion detection framework for resource-constrained edge environments, balancing detection performance, reliability, and inference efficiency.
- Built a leakage-aware evaluation pipeline and investigated tabular-to-image representation learning, deep-learning baselines, and deployment-oriented model optimization.

Education

Hanoi University of Industry

Bachelor of Computer Science

GPA: 3.76/4.00 | Class Valedictorian

2022 – 2026

Technical Skills

ProgrammingPython, C/C++, SQL, JavaScript, C#

LLM & Agentic AI LLM Agents, Multi-step Workflows, Prompt Engineering, Structured Outputs, RAG, Retrieval Orchestration, LLM Evaluation, LLM Observability, Dify, Langfuse

Machine LearningPyTorch, Hugging Face, Scikit-learn, OpenCV, NumPy, Pandas, Deep Learning, Multimodal Learning, Audio Processing, Computer Vision

AI EngineeringREST APIs, Docker, Linux, Git, On-premise Integration, Experiment Tracking, Model Evaluation, Inference Pipelines

CybersecuritySOC Tier 1, SIEM, SOAR, EDR, Intrusion Detection, Alert Triage, Process-tree Analysis, Command-line Analysis

Selected Awards & Achievements

- Class Valedictorian, Bachelor of Computer Science, Hanoi University of Industry2026
- First Prize, University Scientific Research Competition2026
- Second Prize, Digital Era Creative Innovation Competition2026
- Second Prize, AI Startup Competition2026
- First Prize, AI Assistant Competition2025

Languages

English

Japanese

TOEIC 670

JLPT N3